

Django Project Structure & Commands

Command: `python -m venv venv`

Purpose: Creates a virtual environment (venv folder). Keeps project packages isolated.

Command: `venv\Scripts\activate`

Purpose: Activates the virtual environment (Windows).

Command: `pip install django`

Purpose: Installs Django.

Command: `django-admin startproject newproject`

Purpose: Creates project folder with `manage.py` and inner `newproject` package.

Command: `python manage.py startapp newapp`

Purpose: Creates app folder `newapp` with `admin.py`, `apps.py`, `models.py`, `views.py`, etc.

Command: `python manage.py runserver`

Purpose: Starts development server at `http://127.0.0.1:8000/`.

Command: `python manage.py makemigrations`

Purpose: Creates migration files in `migrations/` after model changes.

Command: `python manage.py migrate`

Purpose: Applies migrations to database (`db.sqlite3`).

Command: `python manage.py createsuperuser`

Purpose: Creates Django admin user.

Project Files

manage.py: Runs Django management commands.

newproject/settings.py: Project settings: installed apps, database, middleware, templates.

newproject/urls.py: Maps URLs to views or app URLs.

newproject/asgi.py: ASGI configuration for async deployment.

newproject/wsgi.py: WSGI configuration for deployment.

newapp/views.py: Contains request handling logic and returns responses.

newapp/models.py: Defines database tables/models.

newapp/admin.py: Registers models in Django admin.

newapp/apps.py: App configuration.

newapp/tests.py: Application tests.

newapp/migrations/: Stores migration files.

db.sqlite3: SQLite database file.

venv/: Virtual environment with installed packages.